

Assignment 3 – Containerize Snakemake and Deploy on AWS Batch

This assignment requires you to containerize the genomic analysis pipeline Snakemake script from Assignment 2, build a Docker image, push it to DockerHub, and deploy the containerized pipeline using AWS Batch.

You will set up a managed compute environment, define a Batch job, submit the job, and monitor its execution via CloudWatch logs. This assignment demonstrates the advantages of containerization and managed batch processing for high-throughput bioinformatics workflows

Part A: Build Complete SnakeMake Pipeline

1. Convert the complete genomic analysis bash script to a SnakeMake script. You do not need to include the S3 upload function.
2. Screenshot Requirements:
 - Completed SnakeMake script

Part B: Create a Dockerfile for the Pipeline

1. Write a Dockerfile starting with an official Miniconda base image
2. Install required bioinformatics tools (e.g., BWA, GATK, snpEff, FastQC, samtools)
3. Copy the genomic analysis pipeline SnakeMake script into the container
4. Set the default command to execute the pipeline script
5. Screenshot Requirements:
 - Completed Dockerfile open in VSCode

Part C: Build and Test the Docker Image

1. Build the Docker image on your EC2 server and tag it `FIRST_NAME_assignment_3`
2. Run on your EC2 instance and verify that the pipeline executes successfully
3. Screenshot Requirements:
 - Terminal output of the Docker build process
 - Container execution logs demonstrating successful pipeline completion
 - File listing showing output files generated by the pipeline

Part D: Push Docker Image to DockerHub

1. Tag your local Docker image with your DockerHub repository URI
2. Push the Docker image to DockerHub
3. Screenshot Requirements:
 - Terminal output showing Docker image tagging
 - DockerHub view showing the pushed Docker image details

Part E: Set Up AWS Batch Environment

1. Log in to the AWS Management Console and navigate to AWS Batch

2. Create a managed compute environment
3. Create a job queue and associate it with the compute environment
4. Screenshot Requirements:
 - AWS Batch compute environment configuration screen
 - Job queue configuration page
 - Summary screen showing compute environment and job queue details

Part F: Create AWS Batch Job Definition

1. In AWS Batch, create a new job definition using your Docker image from DockerHub
2. Configure resource parameters (vCPUs, memory) and specify the command to run the pipeline script (e.g., `bash genomic_analysis_script.sh`)
3. Screenshot Requirements:
 - Job definition configuration page showing resource settings and command
 - AWS Batch view showing the job definition details

Part G: Submit and Monitor the AWS Batch Job

1. Submit a job to the AWS Batch queue using your job definition
2. Monitor job progress in the AWS Batch console and view CloudWatch logs for execution details
3. Verify that the pipeline completes successfully and generates the expected output files
4. Screenshot Requirements:
 - AWS Batch job submission details
 - Job status page showing progress and completion
 - CloudWatch logs displaying key pipeline execution messages

Rubric:

Section	Requirements	Marks
Part A	Terminal view displaying the pipeline script	0%
Part B	Completed Dockerfile open in VSCode	25%
Part C	Terminal output of the Docker build process Container execution logs demonstrating successful pipeline completion File listing showing output files generated by the pipeline	15%

Part D	Terminal output showing Docker image tagging DockerHub showing the pushed Docker image details	15%
Part E	AWS Batch compute environment configuration screen Job queue configuration page Summary screen showing compute environment and job queue details	15%
Part F	Job definition configuration page showing resource settings and command Terminal or AWS Batch view showing the job definition details	15%
Part G	AWS Batch job submission details Job status page showing progress and completion CloudWatch logs displaying key pipeline execution messages	15%
Total	PDF file containing all screenshots	100%